MOHAMMAD ULLAH

Department of Electrical and Electronic Engineering (EEE)
Rajshahi University of Engineering & Technology (RUET)

Email: mohammadullah.eee@gmail.com | **Portfolio:** mohammadullah2eee.github.io
LinkedIn: linkedin.com/in/mohammad-ullah-3400123a5

PROFESSIONAL SUMMARY

Electrical and Electronic Engineering student at RUET with hands-on experience in IoT and embedded systems and an undergraduate thesis on photonic integrated circuits (PICs). Motivated to grow into research and engineering roles in which I can design, test and improve technology, with a particular interest in faster and more efficient communication systems. Open to research collaborations, academic inquiries and engineering opportunities.

EDUCATION

B.Sc. in Electrical and Electronic Engineering Batch 2022 – Present
Rajshahi University of Engineering & Technology (RUET)

Higher Secondary Certificate (HSC), Science 2022
Shahid Syed Nazrul Islam College (SSNIC), Mymensingh – GPA 5.00

Secondary School Certificate (SSC), Science 2020
Shohagi Union Higher Secondary School, Mymensingh – GPA 5.00

RESEARCH EXPERIENCE

Undergraduate Thesis Researcher August 2026 – Present
Department of EEE, RUET
Research on photonic integrated circuits using Ansys Lumerical and KLayout.

INTERNSHIPS

Intern, Physical Design – Neural Semiconductor Limited (NSL) 29 Dec 2025 – 02 Jan 2026
Bangladesh-based semiconductor design company
Five-day industrial attachment with technical talks on the overall workflow. Learned the chip design flow, worked on layout using industry-grade technology, and received a certificate of achievement.

Ashuganj Power Station Company Limited (APSCL) Upcoming: February 2027
Internship

PROJECTS

Mach–Zehnder Interferometer (MZI) Design
Designed a silicon photonic MZI layout in KLayout using the SiEPIC PDK. *Tools: KLayout, SiEPIC, Silicon Photonics.* Repository: github.com/mohammadullah2eee/mzi-klayout

Smart Fuel Station Management
ESP32-based vehicle entry/exit and capacity-monitoring prototype with sensors, servo barriers, LCD status, alarms and a web dashboard. *Tools: ESP32, IoT, web dashboard, sensors.*

Smart Cart System
Prototype concept combining load measurement, identification, an LCD and audio feedback to support item verification and inventory tracking. *Tools: embedded systems, load cell, RFID concept.*

Smart Poultry House Monitoring
Low-cost monitoring concept using gas and sound sensing, local alerts and microcontrollers to improve poultry-house awareness. *Tools: ESP32, sensors, IoT.*

Line Following Robot (LFR) with Obstacle Avoidance
Robot that follows a marked line path and detects and avoids obstacles. *Tools: robotics, embedded systems, sensors.*

Home Automation
Home automation system for automatic and convenient control of household appliances. *Tools: embedded systems, IoT.*

LEADERSHIP & ACTIVITIES

Executive Member (Assistant Organizing Secretary), RUET IoT Club Oct 2023 – Aug 2026
Organized weekly sessions and annual events and mentored general members.

Religious Secretary, Greater Mymensingh Association Sep 2026 – Present

General Member, SSNIC Science Club Sep 2020 – Jun 2022

TECHNICAL SKILLS

- **Code & Software:** KLayout, SiEPIC PDK, MZI layout design, Ansys Lumerical, C/C++ for Arduino, Python
- **Circuit & Simulation:** MATLAB, MATLAB Simulink, Visio
- **Embedded & IoT:** ESP32/Arduino, sensor integration, Wi-Fi and web servers, microcontroller projects
- **PCB Design:** EasyEDA
- **VLSI Circuit Design:** Microwind, KLayout
- **Electronics:** Analog and digital circuits, measurements and testing, LCD / I^2C interfaces, prototyping and debugging
- **Research & Collaboration:** Technical documentation, team projects, presentation and communication

AWARDS & RECOGNITION

Best Executive of the Year 2025 2025
RUET IoT Club – awarded for discipline in mentorship and management

2nd Prize, Ramadan Quran Challenge 2025
Organized by RUET

Regional Winner, Bangladesh Mathematical Olympiad 2022

Regional Winner, Bangladesh Mathematical Olympiad 2021

Divisional Winner, Srijonshil Medha Onneson (SMO) 2021
Awarded for successfully passing the Olympiad

INTERESTS

IoT and embedded systems; photonic integrated circuits; PCB design; research and innovation; open source and community.

DECLARATION

I hereby declare that the information provided in this Curriculum Vitae is true and correct to the best of my knowledge.

Mohammad Ullah
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